

Lab - 11

1. Apply Naive Bayes classification for the given database:-

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

$X' = (\text{Outlook} = \text{Sunny}, \text{Temperature} = \text{cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{Strong})?$

is it Yes, or No? Find it using Naive Bayes Classification.

$$\rightarrow P(\text{Play} = \text{Yes}) = 9/14$$

$$P(\text{Play} = \text{No}) = 5/14$$

Outlook	Yes	No	Temp.	Yes	No
Sunny	2/9	3/5	Hot	2/9	2/5
Overcast	4/9	0/5	Mild	4/9	2/5
Rain	3/9	2/5	Cool	3/9	1/5

Humidity	Yes	No	Wind	Yes	No
High	3/9	4/5	Strong	3/9	3/5
Normal	6/9	1/5	Weak	6/9	2/5

$x' = (\text{outlook} = \text{sunny}, \text{temperature} = \text{cool}, \text{Humidity} = \text{High}, \text{Wind} = \text{strong}). = ?$

$$P(c_i | x) = P(x_1 | c_i) \times P(x_2 | c_i) \times \dots \times P(x_n | c_i) \times P(c_i)$$

$$P(\text{Yes} | x') = [P(\text{sunny} | \text{Yes}) \times P(\text{cool} | \text{Yes}) \times P(\text{High} | \text{Yes}) \times P(\text{strong} | \text{Yes})] \times P(\text{Play} = \text{Yes})$$

$$\Rightarrow [2/9 \times 3/9 \times 3/9 \times 3/9] \times 9/14 = 0.0053$$

$$P(\text{No} | x') = [P(\text{sunny} | \text{No}) \times P(\text{cool} | \text{No}) \times P(\text{High} | \text{No}) \times P(\text{strong} | \text{No})] \times P(\text{Play} = \text{No})$$

$$\Rightarrow [3/5 \times 1/5 \times 4/5 \times 3/5] \times 5/14 = 0.0206$$

We label x' to be "No".

2. Apply Gini Index for the given dataset.

RID	Age	Income	Student	Credit rating	Buys Computer
1	Youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle-aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle-aged	low	excellent	excellent	yes
8	youth	medium	fair	fair	no
9	youth	low	fair	fair	yes
10	senior	medium	fair	fair	yes
11	youth	medium	excellent	excellent	yes
12	middle-aged	medium	excellent	excellent	yes
13	middle-aged	high	fair	fair	yes
14	senior	medium	excellent	excellent	no

$$\rightarrow P(\text{buy-Computer} = \text{yes}) = 9/14$$

$$P(\text{buy-computer} = \text{no}) = 5/14$$

$$\text{Gini}(D) = 1 - \left(\frac{9}{14}\right)^2 - \left(\frac{5}{14}\right)^2$$

$$= 0.4592$$

Consider each of possible splitting subsets for Age attribute,

1. {youth, middle, senior}

2. {youth, senior}, {middle}

3. {middle, senior}, {youth}

Distinct Value	Yes	No	Total
youth	2	3	5
middle	4	0	4
senior	3	2	5

Consider subset {youth, middle}

Gini Age $\in$ {youth, middle} (D) =

$$\frac{9}{14} \left(1 - \left(\frac{6}{9} \right)^2 - \left(\frac{3}{9} \right)^2 \right) + \frac{5}{14} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right)$$

$$\Rightarrow 0.4571$$

Gini Age $\in$ {youth, senior} (D) =

$$\frac{10}{14} \left(1 - \left(\frac{5}{10} \right)^2 - \left(\frac{5}{10} \right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{4}{4} \right)^2 - \left(\frac{0}{4} \right)^2 \right)$$

$$\Rightarrow 0.3571$$

Gini Age $\in$ {middle, senior} (D) =

$$\frac{9}{14} \left(1 - \left(\frac{7}{9} \right)^2 - \left(\frac{2}{9} \right)^2 \right) + \frac{5}{14} \left(1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \right)$$

$$\Rightarrow 0.3937$$

Similarity For Income,

1. {high, low}, {medium}

2. {high, medium}, {low}

3. {low, medium}, {high}

Distinct Value	Yes	No	Total
High	2	2	4
Low	3	1	4
medium	4	2	6

Gini Income $\in \{high, low\} (D) =$

$$\frac{8}{14} \left(1 - \left(\frac{5}{8} \right)^2 - \left(\frac{3}{8} \right)^2 \right) + \frac{6}{14} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right)$$

$$\Rightarrow 0.4583$$

Gini Income $\in \{high, medium\} (D) =$

$$\frac{10}{14} \left(1 - \left(\frac{6}{10} \right)^2 - \left(\frac{4}{10} \right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right)$$

$$\Rightarrow 0.4500$$

Gini Income $\in \{low, medium\} (D) =$

$$\frac{10}{14} \left(1 - \left(\frac{7}{10} \right)^2 - \left(\frac{3}{10} \right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right)$$

$$\Rightarrow 0.4429$$

Pon student Attribute

Distinct Value	Yes	No	Total
Yes	6	1	7
No	3	4	7

Gini student(D) = 8

$$\frac{7}{14} \left(1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2 \right) + \frac{7}{14} \left(1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2 \right)$$

$$\Rightarrow 0.3673$$

For Credit rating

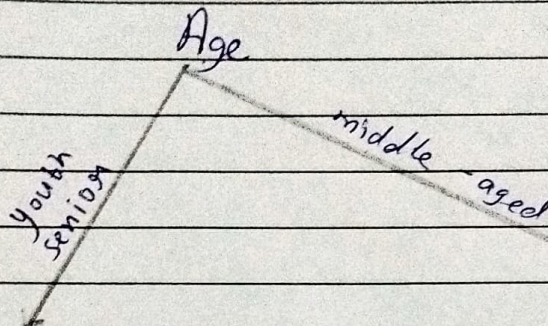
Distinct value	Yes	No	Total
Fair	6	2	8
excellent	3	3	6

Gini Credit rating (D) =

$$\frac{8}{14} \left(1 - \left(\frac{6}{8}\right)^2 - \left(\frac{2}{8}\right)^2 \right) + \frac{6}{14} \left(1 - \left(\frac{3}{6}\right)^2 - \left(\frac{3}{6}\right)^2 \right)$$

$$\Rightarrow 0.4286$$

Attribute	Gini Index	Δgini
Age	0.3571	$0.4592 - 0.3571 = 0.1021$
Income	0.4429	$0.4592 - 0.4429 = 0.0163$
Student	0.3673	$0.4592 - 0.3673 = 0.0919$
Credit rating	0.4286	$0.4592 - 0.4286 = 0.0306$



In come	Student	CR	class
high	no	fair	no
high	no	ex.	no
medium	no	fair	no
low	yes	fair	yes
medium	yes	ex.	yes
medium	no	fair	yes
low	yes	fair	yes
low	yes	ex.	no
medium	yes	fair	yes
medium	no	ex	no

Income	Student	CR	class
high	no	fair	yes
low	yes	ex.	yes
medium	no	ex.	yes
high	yes	fair	yes